NUMERO DI NEPERO $e = 2,718281828846...$

FUNZIONE ESPONENZIALE

$$y = a^x \qquad a > 0 \; ; \; a \neq 1$$

$$y = e^x$$

Rendimento bancario del 100% dopo 1 anno.

C = Capitale (1000). Otteniamo il montante M.
r = rendimento$_{anno}$ (100%)

$$M = 1000 + 1000 \cdot 100\% = 1000\left(1 + 100\%\right) = 1000 \cdot 2 = 2000$$

$$M = C\left(1 + r\right)$$

100% → 1
50% → 0,5
25% → 0,25
1% → 0,01

Rendimento bancario del 50% qui 6 mesi per l'anno.

$$M_{1/2} = \left(1000 + 1000 \cdot 50\%\right) = 1500. \quad 1° \text{ semestre.}$$

$$M_1 = \left(1500 + 1500 \cdot 50\%\right) = 2250 \quad 2° \text{ semestre.}$$

$$M_{TOT} = \left(1000 + 1000 \cdot 50\%\right) + \left(1000 + 1000 \cdot 50\%\right) \cdot 50\% =$$

$$= \left(1000 + 1000 \cdot 50\%\right)\left(1 + 50\%\right) =$$

$$= 1000\left(1 + 50\%\right)\left(1 + 50\%\right) = 1000\left(1 + 50\%\right)^2 =$$

$$M = C\left(1 + \frac{r}{2}\right)^2$$

Rendimento valutato ogni 4 mesi (3 periodi)

$$M = C\left(1 + \frac{r}{3}\right)^3$$

$$M = 1000\left(1 + \frac{100\%}{3}\right)^3 = 2370$$

Interesse calcolato giornalmente

$$M = 1000\left(1 + \frac{100\%}{365}\right)^{365} = 2'714$$

2,714

$e = \left(1 + \frac{1}{n}\right)^n$ per $n \to +\infty$

PROPRIETÀ DELLE POTENZE

$a^n = \underbrace{a \cdot a \cdot \ldots \cdot a}_{n \text{ volte}}$

$a^m + a^n =$

$3^6 + 3^4 = 3^4 \cdot 3^2 + 3^4 = 3^4 (3^2 + 1) = 10 \cdot 3^4$

$9^3 + 3^2 = (3^2)^3 + 3^2 = 3^6 + 3^2 = 3^2 \cdot 3^4 + 3^2 = 3^2 (3^4 + 1) = 82 \cdot 3^2$

$3^4 + 5^2$ si lascia così.

- $a^n \cdot a^m = a^{n+m}$
- $a^n \cdot b^n = (a \cdot b)^n$ } PRODOTTO
- $a^n : a^m = a^{n-m}$
- $(a^m)^n = \underbrace{a^m \cdot a^m \cdot \ldots \cdot a^m}_{n \text{ volte}} = a^{\overbrace{m+m+\ldots m}^{m \text{ volte}}} = a^{m \cdot n}$

- $a^{-m} = \frac{1}{a^m}$
- $a^0 = 1$

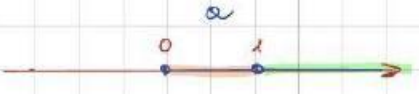
$a^0 = a^{3-3} = a^3 : a^3 = 1$

- $a^{\frac{1}{m}} = \sqrt[m]{a}$; $a^{\frac{1}{2}} = \sqrt{a}$
- $a^{\frac{m}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$
- $\left(\frac{a}{b}\right)^{-\frac{m}{n}} = \left(\frac{b}{a}\right)^{\frac{m}{n}} = \sqrt[n]{\left(\frac{b}{a}\right)^m}$

FUNZIONE ESPONENZIALE

Si dice FUNZIONE ESPONENZIALE elementare di BASE a, con a numero positivo diverso da 1, la seguente equazione:

$$y = a^x \quad ; \quad a > 0 \wedge a \neq 1$$



Es: $y = 3^x$; $y = 2^x$; $y = e^x$; $y = \left(\frac{1}{2}\right)^x$

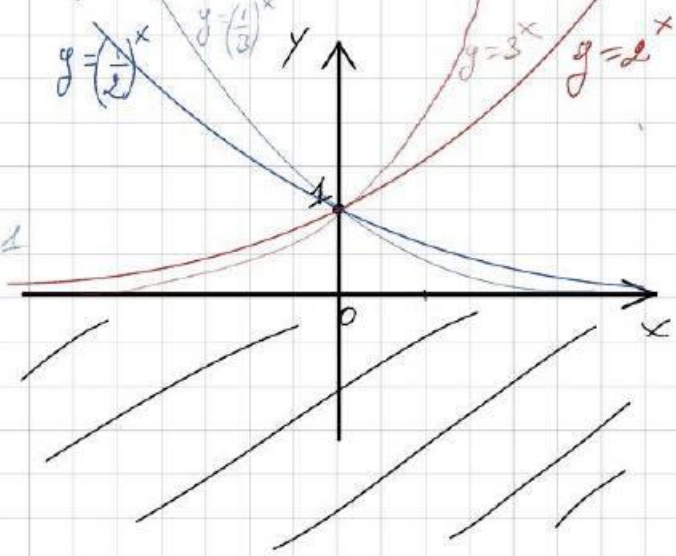
GRAFICO DECCA FUNZ. ESPON. ELEMENTARE

$y = a^x$

$y = 2^x$; $a > 1$

$y = \left(\frac{1}{2}\right)^x$; $0 < a < 1$

$\left(\frac{1}{2}\right)^x = 2^{-x}$

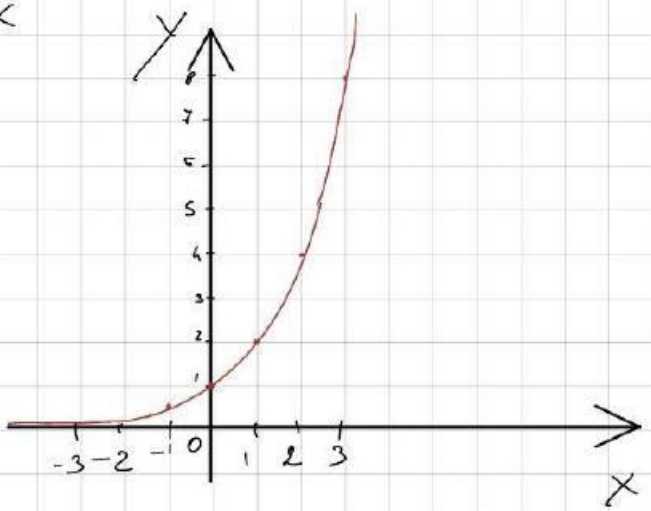


La funzione esponenziale è sempre positiva $a^x > 0$, infatti passerà per il punto $(0,1)$. Il dominio è $\forall x \in \mathbb{R}$ l'asse X è l'asintoto per la funzione.

Se $a > 1$ la funzione è strettamente crescente.

Se $0 < a < 1$ " " " " decrescente.

Rappresentiamo $y = 2^x$



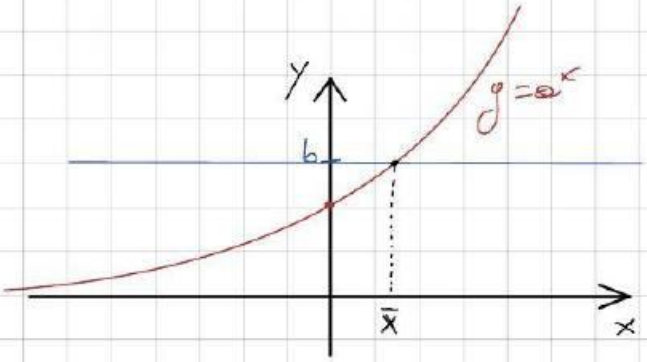
x	y
-2	$2^{-2} = \frac{1}{4}$
-1	$2^{-1} = 1/2$
0	$2^0 = 1$
1	$2^1 = 2$
2	$2^2 = 4$
3	$2^3 = 8$

EQUAZIONI ESPONENZIALI

$$a^x = b$$

- Se $b \leq 0 \rightarrow a^x = b$ è impossibile.
- Se $b > 0 \rightarrow a^x = b$ ha una e una sola soluzione (la funzione esponenziale è sempre MONOTONA)

$$\begin{cases} y = a^x \\ y = b \end{cases}$$



Per il confronto:

$a^x = b$

- $2^x = 16$

Dobbiamo porre 1° e 2° membro sotto la stessa base

$2^x = 2^4$

Due potenze sono uguali se hanno la stessa base e lo stesso esponente.

$\boxed{x = 4}$

- $2^{3x-5} = 8$

$2^{3x-5} = 2^3$

$3x - 5 = 3$

$3x = 8$

$x = \frac{8}{3}$

- $16^x = 2\sqrt{2}$

$(2^4)^x = 2 \cdot 2^{1/2}$

$2^{4x} = 2^{1+\frac{1}{2}}$

$4x = \frac{3}{2} \qquad x = \frac{3}{8}$

n° 68 $\sqrt{3^x \cdot \sqrt{9^{x^2-1}}} = 3^{\frac{x+1}{4}}$

$\left(3^x \cdot \sqrt{9^{x^2-1}}\right)^{\frac{1}{2}} = 3^{\frac{x+1}{4}}$

$\left(3^x \cdot (9^{x^2-1})^{\frac{1}{2}}\right)^{\frac{1}{2}} = 3^{\frac{x+1}{4}}$

$(3^x)^{\frac{1}{2}} \cdot \left((9^{x^2-1})^{\frac{1}{2}}\right)^{\frac{1}{2}} = 3^{\frac{x+1}{4}}$

$3^{\frac{x}{2}} \cdot (9^{x^2-1})^{\frac{1}{4}} = 3^{\frac{x+1}{4}}$

$3^{\frac{x}{2}} \cdot 3^{\frac{2(x^2-1)}{4}} = 3^{\frac{x+1}{4}}$

$3^{\frac{x}{2}} \cdot 3^{2 \cdot \frac{x^2-1}{4}} = 3^{\frac{x+1}{4}}$

$3^{\frac{x}{2}} \cdot 3^{\frac{x^2-1}{2}} = 3^{\frac{x+1}{4}}$

$3^{\frac{x}{2} + \frac{x^2-1}{2}} = 3^{\frac{x+1}{4}}$

$\frac{x}{2} + \frac{x^2-1}{2} = \frac{x+1}{4}$

$2x + 2x^2 - 2 = x + 1$

$2x^2 + x - 3 = 0$

$x_{1,2} = \frac{-1 \pm \sqrt{1+24}}{4} = \frac{-1 \pm 5}{4}$ → $-\frac{3}{2}$, 1

n. 69

$\sqrt{e^{-|x+1|}} = \frac{e^{|x+1|}}{e^3 \sqrt{e}}$

$e^{\frac{-|x+1|}{2}} = \frac{e^{|x+1|}}{e^3 \cdot e^{1/2}}$

$e^{\frac{-|x+1|}{2}} = \frac{e^{|x+1|}}{e^{7/2}}$

$2 \cdot \left(-\frac{|x+1|}{2}\right) = 2 \cdot \left(|x+1| - \frac{7}{2}\right)$

$-|x+1| = 2|x+1| - 7$

$-|x+1| - 2|x+1| = -7$

$-3|x+1| = -7$

$3|x+1| = 7$

$|x+1| = \begin{cases} x+1 & \text{se } x \geq -1 \\ -x-1 & \text{se } x < -1 \end{cases}$

Se $x \geq -1$ $\quad 3x + 3 = 7$

$x = \frac{4}{3}$

Se $x < -1$ $\quad -3x - 3 = 7$

$-3x = 7 + 3$

$-3x = 10$

$x = -\frac{10}{3}$

Riduzione mediante variabile ausiliaria

n°85 pg 793

$2^{2x} - 7 \cdot 2^x - 8 = 0$ $\quad \left((2^x)^2 = 2^{2x} \right)$

$t = 2^x$

$t^2 - 7t - 8 = 0$

$(t+1)\cdot(t-8) = 0$ $\quad \begin{cases} t+1=0 \; ; \; t=-1 \\ T-8=0 \; ; \; T=8 \end{cases}$

$t = \boxed{-1 = 2^x}$ impossibile

$t = \boxed{8 = 2^x} \rightarrow 2^3 = 2^x \; ; \; x = 3$

n°89

$\frac{1}{2^x} = 2^x - \frac{3}{2}$ $\quad 2^x = t$

$2t \frac{1}{t} = \left(t - \frac{3}{2}\right) \cdot 2t$

$2 = 2t^2 - 3t$

$2t^2 - 3t - 2 = 0$

$t_{1,2} = \frac{3 \pm \sqrt{9+16}}{4} = \frac{3 \pm 5}{4}$ $\quad \begin{cases} 2 = 2^x \; ; \; x = 1 \\ -\frac{1}{2} = 2^x \; ; \; \text{imp.} \end{cases}$

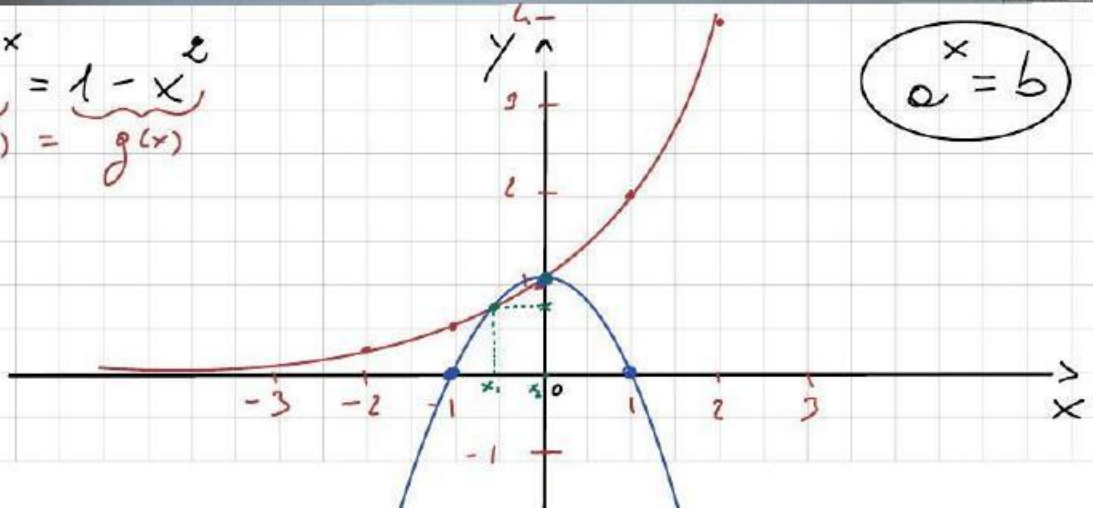
ES. ESPONENZIALI RISOLUBILI GRAFICAMENTE

ESERCIZIO GUIDATO

Interpretando graficamente l'equazione $2^x = 1 - x^2$, stabilisci il numero delle sue soluzioni e, per ciascuna di esse, determina un intervallo di lunghezza unitaria cui essa appartiene.

$2^x = 1 - x^2$

$f(x) = g(x)$

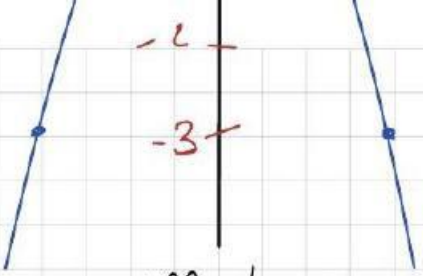


$a^x = b$

x	y
-2	$2^{-2} = 1/4$
-1	$2^{-1} = 1/2$
0	$2^0 = 1$
1	$2^1 = 2$
2	$2^2 = 4$

2^x

$1 - x^2$

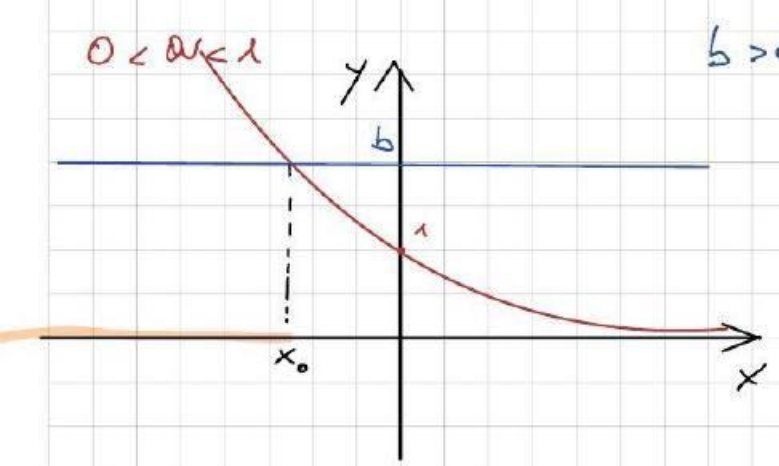


x	y
-3	$1-(-3)^2=-8$
-2	$1-(-2)^2=-3$
-1	$1-(-1)^2=0$
0	$1-(0)^2=1$
1	$1-(1)^2=0$
2	$1-(2)^2=-3$
3	$1-(3)^2=$

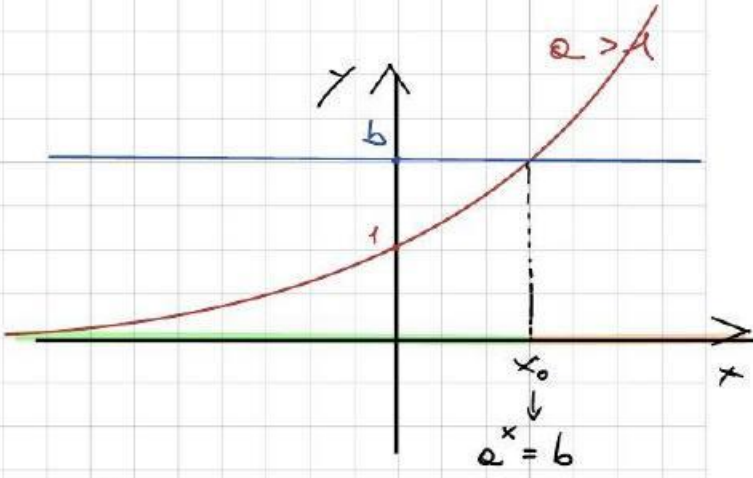
Le soluzioni sono 2, una delle due è nota e vale "0" mentre l'altra cade nell'intervallo $[-1,0]$.

DISEQUAZIONI ESPONENZIALI

una disequazione si dice esponenziale quando l'incognita compare nell'esponente di almeno una potenza



$b > 0$



- $a^x > b$

Risolvo l'eq. associata $a^x = b$

Trovo x_0 e la disq. avrà come soluzione $x < x_0$

- $a^x < b$

Risolvo l'eq. associata $a^x = b$

Trovo x_0 e la disq. avrà come soluzione $x > x_0$

- $a^x \geq b$

Risolvo l'eq. associata $a^x = b$

Trovo x_0 e la disq. avrà come soluzione $x \geq x_0$

- $a^x \leq b$

Risolvo l'eq. associata $a^x = b$

Trovo x_0 e la disq. avrà come soluzione $x \leq x_0$

Esempio:

$2^x > 16$

(Risolvo $2^x = 16$; $2^x = 2^4$; $x = 4$ (x_0))

S: $x > 4$

$3^x < 27$

$3^{x} < 3^{3}$

$x < 3$

- $\left(\frac{3}{4}\right)^x \geq \frac{16}{9}$

$\left(\frac{3}{4}\right)^x \geq \left(\frac{4}{3}\right)^2$; $\left(\frac{3}{4}\right)^x \geq \left(\frac{3}{4}\right)^{-2}$

<1

$x \leq -2$

- pg 805 n° 387.

$3^{3x} - 3^{x+1} - 3^{2x+1} + 3 \geq 0$

$3^{3x} - 3 \cdot 3^{2x} - 3 \cdot 3^{x} + 3 \geq 0$

t^3 t^2 t

3^x 3^{2x} 3^{3x}

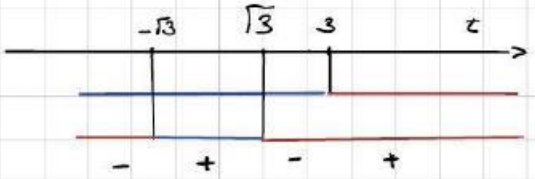
$t^3 - 3t^2 - 3t + 3 \geq 0$

$t^2(t-3) - 3(t-3) \geq 0$

$(t-3)(t^2-3) \geq 0$

$t - 3 \geq 0 \quad t \geq 3$

$t^2 - 3 \geq 0 \quad t \leq -\sqrt{3} \vee t \geq \sqrt{3}$



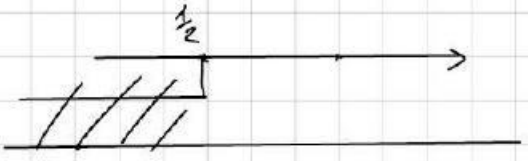
$-\sqrt{3} \leq t \leq \sqrt{3} \quad \vee \quad t \geq 3$

$-\sqrt{3} \leq 3^x \leq \sqrt{3} \quad \vee \quad 3^x \geq 3$

① $3^x \geq -\sqrt{3}$ $\forall x \in \mathbb{R}$

② $3^x \leq \sqrt{3}$; $3^x \leq 3^{1/2}$; $x \leq \frac{1}{2}$

③ $3^x \geq 3$; $x \geq 1$; $x \geq 1$



S: $x \leq \frac{1}{2} \vee x \geq 1$



321 $2^x - 10\sqrt{2^x} + 16 \leq 0$ $[2 \leq x \leq 6]$

t^2 t

$t^2 - 10t + 16 \leq 0$

$(t-2)(t-8) \leq 0$

$(t-2)(t-8) = 0 \quad t=2 \quad t=8$

$2 \leq t \leq 8$

$$\begin{cases} t \geq 2 \\ t \leq 8 \end{cases} \quad \begin{cases} \sqrt{2^x} \geq 2 \\ \sqrt{2^x} \leq 8 \end{cases} \quad \begin{cases} 2^{\frac{x}{2}} \geq 2 \\ 2^{\frac{x}{2}} \leq 8 \end{cases} \quad \begin{cases} 2^{\frac{x}{2}} \geq 2^1 \\ 2^{\frac{x}{2}} \leq 2^3 \end{cases} \Rightarrow \frac{x}{2} \geq 1 \Rightarrow x \geq 2$$

$$\frac{x}{2} \leq 3 \Rightarrow x \leq 6$$

$$2 \leq x \leq 6$$

322 $3^{\frac{2}{x}} \geq 12 \cdot 3^{\frac{1}{x}} - 27$ $\quad \left[x < 0 \vee 0 < x \leq \frac{1}{2} \vee x \geq 1\right]$

$3^{\frac{1}{x}} = \left(3^{\frac{1}{x}}\right)^2$ $\quad$ C.E. $x \neq 0$

$t^2 \geq 12t - 27$

$t^2 - 12t + 27 \geq 0$ $\quad t \leq 3 \vee t \geq 9$

$(t-3)(t-9) \geq 0$

$(t-3)(t-9) = 0 \begin{cases} t = 3 \\ t = 9 \end{cases}$

$t \leq 3$ $\quad$ $t \geq 9$

$3^{\frac{1}{x}} \leq 3^1$ $\quad \cup \quad$ $3^{\frac{1}{x}} \geq 9$ $\quad (3^2)$

$\frac{1}{x} \leq 1$ $\quad$ $\frac{1}{x} \geq 2$

$x \geq 1$ $\quad \cup \quad$ $x \leq \frac{1}{2}$



$x < 0 \vee 0 < x \leq \frac{1}{2} \vee x \geq 1$

331 $\frac{e^{-3x} - \sqrt{e^{2x-1}}}{e^{2x}} > 0$ $\quad \left[x < \frac{1}{8}\right]$

$\frac{N(x)}{D(x)} > 0$ $\quad$ $N(x) > 0$, $D(x) > 0$

$N: e^{-3x} - \sqrt{e^{2x-1}} > 0$ $\quad D: e^{2x} > 0 \quad \forall x \in \mathbb{R}$

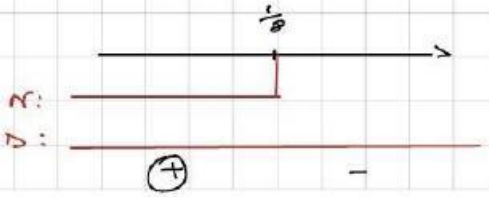
$e^{-3x} - e^{\frac{2x-1}{2}} > 0$

$e^{-3x} > e^{\frac{2x-1}{2}}$

$-3x > \frac{2x-1}{2}$

$-6x - 2x + 1 > 0$

$8x < 1 \quad x < \frac{1}{8}$



$S: \quad x < \frac{1}{8}$

FUNZIONE LOGARITMO

Dati 2 numeri reali positivi a e b con a diversa da 1, si chiama Logaritmo di base a del numero b, l'esponente al quale si deve elevare la base a per ottenere b.

$$x = \log_a b \iff a^x = b$$

(b: argomento; a: base del logaritmo)

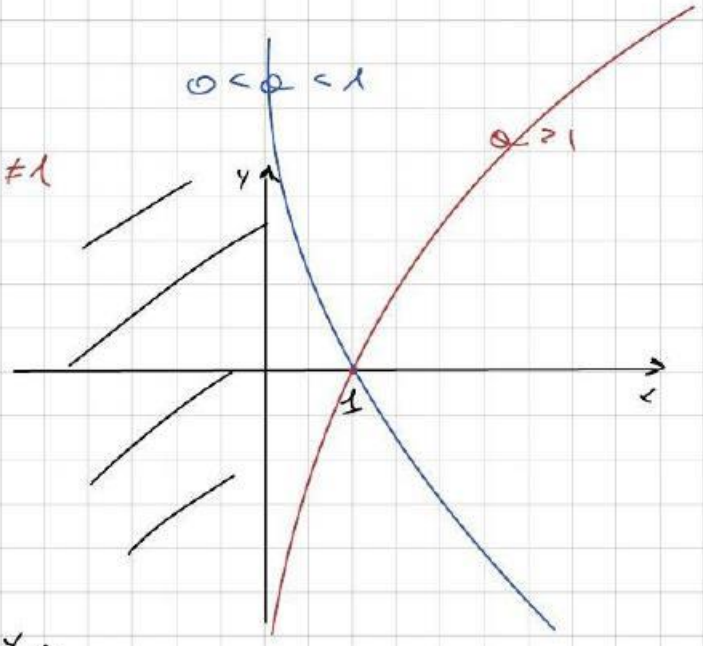
$\log_2 8 = 3$ $\quad 2^{(3)} = 8$

$\log_3 1 = 0$ $\quad 3^{(0)} = 1$

$\log_4 256 = 4$ $\quad 4^{(4)} = 256$

$\log_{\frac{1}{3}} 27 = -3$ $\quad \left(\frac{1}{3}\right)^{(-3)} = 27$

$y = \log_a x \qquad a > 0 \wedge a \neq 1$

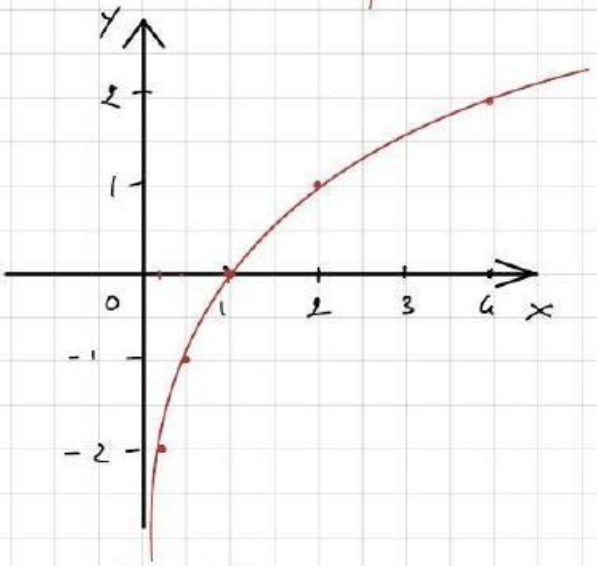


Rappresentiamo

$y = \log_2 x$

$2^{-2} = \frac{1}{4}$

x	y
1/4	$\log_2 \frac{1}{4} = -2$
1/2	$\log_2 \frac{1}{2} = -1$
1	$\log_2 1 = 0$
2	$\log_2 2 = 1$
4	$\log_2 4 = 2$



Il dominio della funzione logaritmo è $x > 0$ $(\mathbb{R}^+)$

$3 = \log_2 2^3$

$3 = 3 \cdot \log_2 2$

$\log_2 2 = 1$

PROPRIETA' DEI LOGARITMI

- $\log_a (b \cdot c) = \log_a b + \log_a c$
- $\log_a b^m = m \cdot \log_a b$
- $\log_a \frac{b}{c} = \log_a b - \log_a c$
- $\log_a b = \frac{\log_c b}{\log_c a}$; Es: $\log_2 7 = \frac{\log_{10} 7}{\log_{10} 2} = 2,81$

- Dimostrazione

$\log_a (b \cdot c) = \log_a b + \log_a c$

x $\quad$ y

$x = \log_a b \iff a^x = b$

$y = \log_a c \iff a^y = c$ $\quad$ Moltiplichiamo m. a. m.

$a^x \cdot a^y = b \cdot c$

$a^{x+y} = b \cdot c$

$\log_a (b \cdot c) = x + y$ $\quad$ sost. x e y.

$\log_a (b \cdot c) = \log_a b + \log_a c$ $\quad$ c.v.d.

Ultime conseguenze

- $\log_a b = \frac{\log_b b}{\log_b a} = \frac{1}{\log_b a} = (\log_b a)^{-1}$
- $\log_{\frac{1}{a}} b = \frac{\log_a b}{\log_a \frac{1}{a}} = \frac{\log_a b}{-\log_a a} = -\log_a b$

Es pg 760 n° 283

$\log_2 3 + \log_2 6 = \log_2 (3 \cdot 6) = \log_2 18$

n° 302

$1 + \log \sqrt{10} - \frac{1}{3} \cdot \log 10 - \log \sqrt[6]{10}$

$1 + \log 10^{\frac{1}{2}} - \frac{1}{3} \log 10 - \log 10^{\frac{1}{6}}$

$1 + \frac{1}{2} \cdot \log 10 - \frac{1}{3} \log 10 - \frac{1}{6} \log 10 =$

$1 + \log_{10} 10 \cdot \left(\frac{1}{2} - \frac{1}{3} - \frac{1}{6}\right) = 1 + \frac{3-2-1}{6} = 1 = \log 10$

Es. pag 760 da n° 280 a 285, 301, 303, 312, 313, 324, 325, 323, 340, 341, 344

364 $\left(\frac{2\ln\sqrt{\pi} - \log_\pi e}{\ln\pi - 1} - 1\right) \cdot \ln\pi^2$ [2]

$2 \ln \sqrt{\pi}$

$2 \ln \pi^{\frac{1}{2}} = 2 \cdot \frac{1}{2} \cdot \ln \pi$

$\left(\frac{\ln \pi - \log_\pi e}{\ln \pi - 1} - 1\right) \cdot 2 \ln \pi$

$\log_\pi e = \frac{\ln e}{\ln \pi} = \frac{1}{\ln \pi} = \ln^{-1} \pi$

$\left(\frac{\ln \pi - \frac{1}{\ln \pi}}{\ln \pi - 1} - 1\right) 2 \ln \pi$

$\left(\frac{\frac{\ln^2 \pi - 1}{\ln \pi}}{\ln \pi - 1} - 1\right) 2 \ln \pi$

$\left(\frac{\ln^2 \pi - 1}{\ln \pi} \cdot \frac{1}{\ln \pi - 1} - 1\right) 2 \ln \pi$

$\left(\frac{(\ln \pi + 1)(\ln \pi - 1)}{\ln \pi \cdot (\ln \pi - 1)} - 1\right) 2 \ln \pi$

$\left(\frac{\ln \pi + 1 - \ln \pi}{\ln \pi}\right) \cdot 2 \ln \pi = 2$ 

EQUAZIONI LOGARITMICHE

DEFINIZIONE Equazione logaritmica

Un'equazione si dice **logaritmica** se l'incognita compare nell'argomento di almeno un logaritmo.

$$\log_a A(x) = \log_a B(x)$$

$$\Downarrow$$

$$A(x) = B(x)$$

Due logaritmi sono uguali se hanno la stessa base e lo stesso argomento.

PRIMA DI RISOLVERE L'EQUAZIONE BISOGNA PORRE LE C.E.

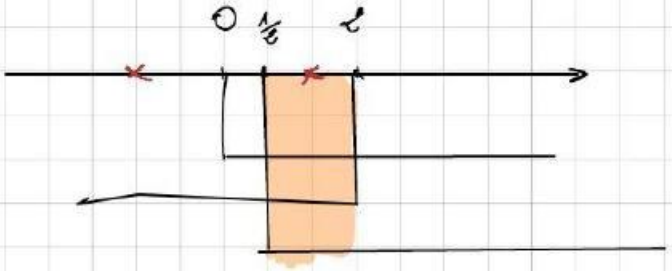
$$\begin{cases} A(x) > 0 \\ B(x) > 0 \end{cases}$$

Risolviamo:

a. $\log_2 x + \log_2 (2 - x) = \log_2 (2x - 1)$

C.E.:

$$\begin{cases} x > 0 \\ 2 - x > 0 \\ 2x - 1 > 0 \end{cases} \qquad \begin{cases} x > 0 \\ x < 2 \\ x > \frac{1}{2} \end{cases}$$



$\frac{1}{2} < x < 2$

$$\log_2 \left(x \cdot (2-x)\right) = \log_2 (2x-1)$$

$$x(2-x) = 2x - 1$$

$$x^2 - 1 = 0 \Rightarrow x = \pm 1$$

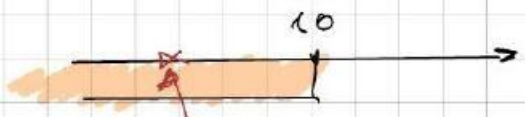
$x = +1$

$x = -1$ non accettabile.

464 $2\log_4(10 - x) = 3$

C.E.:

$10 - x > 0 \Rightarrow x < 10$.



d. a. l. m. x. 2

$$\log_4 (10 - x) = \frac{3}{2}$$

$\log_a a^m = m$

$\log_4(10-x) = \log_4 4^{3/2}$

$4^{3/2} = \left(4^{1/2}\right)^3 = \left(\sqrt{4}\right)^3 = 2^3 = 8$

$10 - x = 8$

$x = 10 - 8 = 2$

$S: x = 2$

$\log_{\frac{1}{2}}\left(\frac{x^2}{2} - 3x - 4\right) = -1 + \log_5 5$ (-1)

$\log_{\frac{1}{2}}\left(\frac{x^2}{2} - 3x - 4\right) = -2$ (-2) $= \log_{\frac{1}{2}}\left(\frac{1}{2}\right)^{-2}$

$\log_a f(x) = b$

$a^b = f(x)$

$\left(\frac{1}{2}\right)^{-2} = \left(\frac{x^2}{2} - 3x - 4\right)$

$4 = \frac{x^2}{2} - 3x - 4$

$x^2 - 6x - 16 = 0$

$x_{1,2} = 3 \mp \sqrt{9+16} = 3 \pm 5$ → 8, -2

C.E.

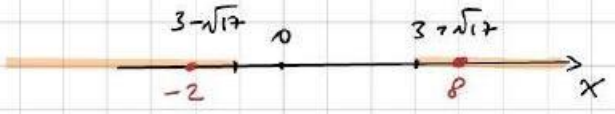
$\frac{x^2}{2} - 3x - 4 > 0$

$x^2 - 6x - 8 > 0$

$x_{1,2} = 3 \mp \sqrt{9+8} = 3 \mp \sqrt{17}$

$x < 3-\sqrt{17} \vee x > 3+\sqrt{17}$

$S: \{-2, 8\}$.



DISEQUAZIONI LOGARITMICHE

Si chiama disequazione logaritmica una disequazione in cui l'incognita compare nell'argomento di almeno un logaritmo

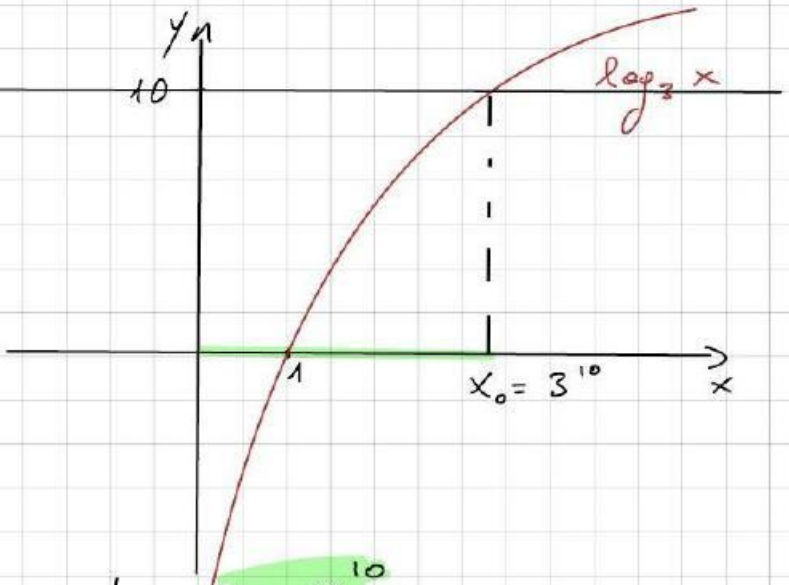
$\log_3 x \leq 10$

C.E. $x > 0$

Risolviamo l'eq. associata

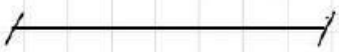
$\log_3 x = 10$

$x = 3^{10}$

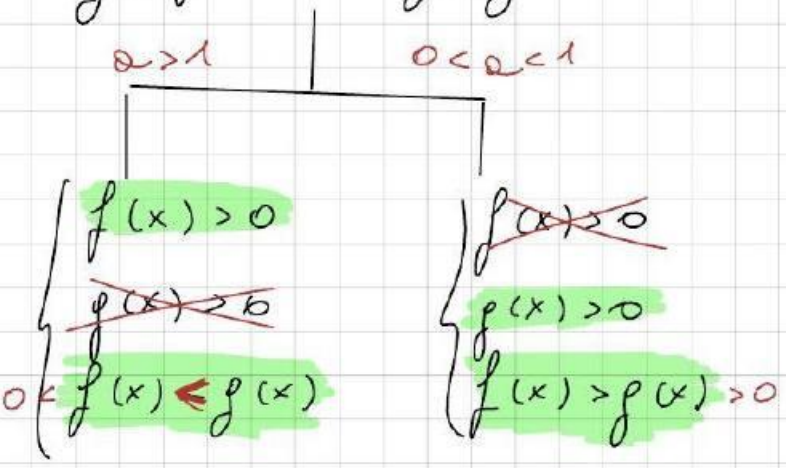


Quindi $\log_3 x \leq 10$ quando $x \leq 3^{10}$

la soluzione sarà $0 < x \leq 3^{10}$



$$\log_a f(x) < \log_a g(x)$$



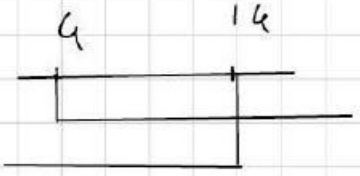
- $\log_a f(x) \leq b$

n°612 pag 815

$\log(x-4) < 1$

$\log(x-4) < \log 10$

C.E. $\begin{cases} x-4 > 0 & x > 4 \\ x-4 < 10 & x < 14 \end{cases}$



S: $4 < x < 14$

- $\log_a f(x) \leq \log_a g(x)$

n°644

$\log_{\frac{1}{3}}(x+1) \leq \log_{\frac{1}{3}}\left(\frac{x}{2}\right)^{-1}$

$\begin{cases} x+1 > 0 \\ \frac{x}{2} > 0 \\ x+1 \geq \frac{2}{x} \end{cases}$

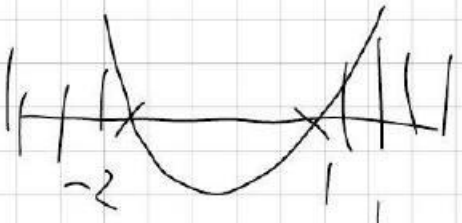
cambio il verso perché $a = \frac{1}{3} < 1$

$$\begin{cases} x > -1 \\ x > 0 \\ x + 1 - \frac{2}{x} \geq 0 \end{cases} \qquad \begin{cases} \frac{x^2 + x - 2}{x} \geq 0 \end{cases}$$

$N \geq 0 \qquad x^2 + x - 2 \geq 0 \qquad x_{12} = \frac{-1 \pm \sqrt{1 - 4(-2)}}{2} =$

$D > 0 \qquad x > 0$

$$\frac{-1 \pm \sqrt{1+8}}{2} = \frac{-1 \pm 3}{2} \begin{cases} \frac{-4}{2} = -2 \\ \frac{2}{2} = 1 \end{cases}$$



$x \leq -2 \vee x \geq 1$

		-2	0	1
D	−	−	+	+
N	+	−	−	+

$$\frac{N}{D} \qquad - \; \oplus \; - \; \oplus$$

681 $\dfrac{1}{\log^2 x+\log x}-\dfrac{2}{\log^2 x-1}>\dfrac{1}{\log^2 x-\log x}$

$[1<x<10 \wedge x\neq 10^{-1}]$

C.E.

$$\begin{cases} x>0 \\ \log^2 x+\log x\neq 0 \\ \log^2 x-1\neq 0 \\ \log^2 x-\log x\neq 0 \end{cases}$$

$$\begin{cases} \log x(\log x+1)\neq 0 \\ (\log x+1)(\log x-1)\neq 0 \\ \log x(\log x-1)\neq 0 \end{cases}$$

$$\begin{cases} \log x\neq 0 & x\neq 1 \\ \log x\neq 1 & x\neq 10 \\ \log x\neq -1 & x\neq 10^{-1} \end{cases}$$

$t=\log x$

$$\frac{1}{t^2+t}-\frac{2}{t^2-1}-\frac{1}{t^2-t}>0$$

$$\frac{1}{t(t+1)}-\frac{2}{(t-1)(t+1)}-\frac{1}{t(t-1)}>0$$

$$\frac{t-1-2t-t-1}{t(t+1)(t-1)}>0$$

$$\frac{-2t-2}{t(t+1)(t-1)}>0$$

$$\frac{-2(t+1)}{t(t+1)(t-1)}>0$$

$$\frac{-2}{t(t-1)}>0$$

$$t(t-1)<0$$

$$0<t<1$$

$$0<\log x<1$$

$$\log 1<\log x<\log 10$$

$$1<x<10$$

EQ ED DISEQ. ESPONENZIALI RISOLVIBILI CON I LOGARITMI

pag 821 n° 475 (EQ)

$2^{1-x}=3^{x+1}$ Applichiamo il logaritmo (base 10) ad e.i.m.

$\log(2^{1-x})=\log(3^{x+1})$

$(1-x)\log 2=(x+1)\cdot\log 3$

$\log 2-x\log 2=x\log 3+\log 3$

$x(-\log 2-\log 3)=\log 3-\log 2$

$x(-\log 6)=\log\frac{3}{2}$

$x=-\dfrac{\log\frac{3}{2}}{\log 6}$

$$x = \frac{\log 3}{2\log 2 - \log 3} = \frac{\log 3}{\log \frac{4}{3}}$$

DISEQ

$$3 \cdot 2^x > 4 \cdot 8^{x+1}$$

$$8 \cdot 2^x > 4 \cdot 8 \cdot 3^x$$

$$2^x > 4 \cdot 3^x$$

$$\frac{2^x}{2^2} > 3^x \quad ; \quad 2^{x-2} > 3^x$$

$$\log 2^{x-2} > \log 3^x$$

$$(x-2)\log 2 > x \log 3$$

$$x \log 2 - 2\log 2 > x \log 3$$

$$x(\log 2 - \log 3) < 2\log 2$$

$$\frac{x(\log 2 - \log 3)}{\log 2 - \log 3} < \frac{2\log 2}{\log 2 - \log 3}$$

$$x < \frac{2\log 2}{\log 2 - \log 3}$$

379 $2^{2x} - 2^{x-1} - 2^{x+2} + 2 < 0$ $\qquad 2^x = t$

$$(2^x)^2 - \frac{2^x}{2} - 4 \cdot 2^x + 2 < 0$$

$$t^2 - \frac{t}{2} - 4t + 2 < 0$$

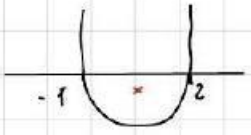
$$t^2 - \frac{9}{2}t + 2 < 0 \quad ; \quad 2t^2 - 9t + 4 < 0$$

$$t_{1,2} = \frac{9 \pm \sqrt{81-32}}{4} = \frac{9 \pm 7}{4}$$

$4^2 = 2^x \Rightarrow x = 2$

$\left(\frac{1}{2}\right)^{-1} = 2^x \Rightarrow x = -1$

$$-1 < x < 2$$



384 $\frac{1}{3} \cdot 3^{x^2-1} \geq \frac{3\sqrt{3}}{3^x}$

$$3^{-1+x^2-1} \geq \frac{3 \cdot 3^{\frac{1}{2}}}{3^x}$$

$$3^{-1+x^2-1} \geq 3^{1+\frac{1}{2}-x}$$

$$x^2 - 2 \geq \frac{3}{2} - x$$

$$x^2 + x - 2 - \frac{3}{2} \geq 0$$

$$x^2 + x - \frac{7}{2} \geq 0$$

$$x_{1,2} = \frac{-1 \pm \sqrt{1+14}}{2} = \frac{-1 \pm \sqrt{15}}{2}$$

$$x \leq \frac{-1-\sqrt{15}}{2} \quad \vee \quad x \geq \frac{-1+\sqrt{15}}{2}$$

956 Considera la funzione $y = \frac{e^{-x} - \sqrt{e}}{e^{3x} - e^{x}}$.

a. Determina il suo dominio.

b. Determina, se esistono, gli zeri della funzione.

c. Studia il segno della funzione e rappresenta le regioni del piano cartesiano dove giace il suo grafico.

[a. $\mathbb{R} - \{0\}$; b. $x = -\frac{1}{2}$; c. $y > 0$ per $-\frac{1}{2} < x < 0$, $y < 0$ per $x < -\frac{1}{2} \vee x > 0$]

$e^{3x} - e^{x} \neq 0 \qquad e^{x} = t$

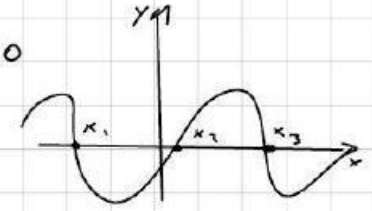
$t^3 - t \neq 0$

$t(t^2 - 1) \neq 0$

$t(t+1)(t-1) \neq 0$

$t \neq 0$	$t \neq 0$	$e^x \neq 0$	$\nexists$
$t+1 \neq 0$	$t \neq -1$	$e^x \neq -1$	$\nexists$
$t-1 \neq 0$	$t \neq 1$	$e^x \neq 1$	$x \neq 0$

D: $\forall x \in \mathbb{R} \setminus \{0\}$



$0 = \frac{e^{-x} - \sqrt{e}}{e^{3x} - e^{x}}$

$e^{-x} - \sqrt{e} = 0 \rightarrow e^{-x} = e^{\frac{1}{2}}$ $\quad x = -\frac{1}{2}$

$\frac{e^{-x} - \sqrt{e}}{e^{3x} - e^{x}} > 0$

N: $e^{-x} - \sqrt{e} > 0$; $e^{-x} > e^{\frac{1}{2}}$; $-x > \frac{1}{2}$; $x < -\frac{1}{2}$



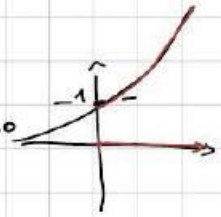
D: $e^{3x} - e^{x} > 0$; $e^{x}(e^{2x} - 1) > 0$; $e^{x}(e^{x}+1)(e^{x}-1) > 0$

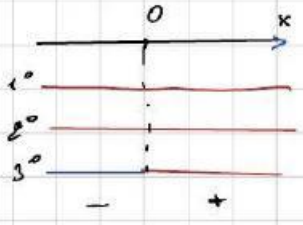
1° 2° 3°

$e^x > 0 \rightarrow \forall x \in \mathbb{R}$

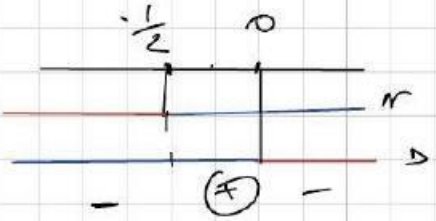
$e^x + 1 > 0 \rightarrow \forall x \in \mathbb{R}$

$e^x - 1 > 0 \rightarrow e^x > 1 \rightarrow x > 0$



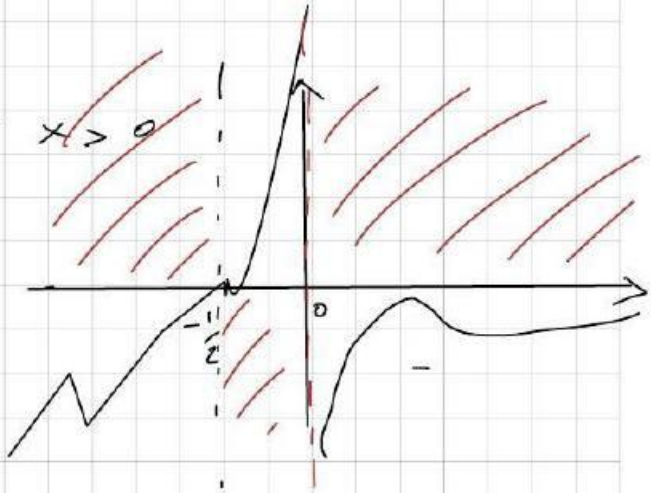


D



La funzione è positiva per $-\frac{1}{2} < x < 0$

è negativa per $x < -\frac{1}{2} \vee x > 0$



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969 Data la funzione $f(x) = (ax^2 + b)e^{cx^2-1}$, determina per quali valori dei parametri a, b, c il suo grafico passa per i punti $\left(0, -\frac{2}{e}\right)$, $(1, 0)$, $(2, 6e^7)$. In seguito determina i punti di intersezione tra la curva trovata e quelle aventi le equazioni seguenti:

a. $y = 4e^{2x^2-1}$

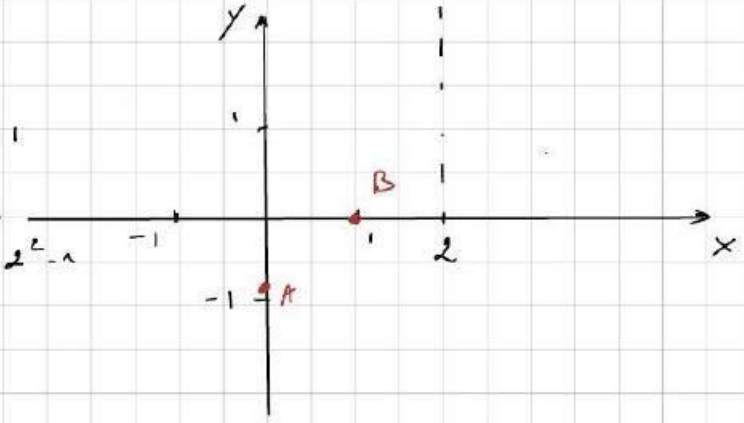
b. $y = 2x^2 - 2$

$\left[a = 2,\ b = -2,\ c = 2;\ \text{a. } (\pm\sqrt{3}, 4e^5);\ \text{b. } (\pm 1, 0), \left(\pm\frac{\sqrt{2}}{2}, -1\right)\right]$

$f(x) = (ax^2 + b)\,e^{cx^2-1}$

Passaggio per A, B, C

$$\begin{cases} -\frac{2}{e} = (a\cdot 0^2 + b)\,e^{c\cdot 0^2-1} \\ 0 = (a\cdot 1^2 + b)\,e^{c\cdot 1^2-1} \\ 6e^7 = (a\cdot 2^2 + b)\,e^{c\cdot 2^2-1} \end{cases}$$



$$\begin{cases} b = -2 \\ 0 = (a-2)\,e^{c-1} \\ 6e^7 = (4a-2)\,e^{4c-1} \end{cases}$$

$$\begin{cases} 4a-2 = 6 \\ 4c-1 = 7 \end{cases} \quad a = 2 \quad c = 2$$

$f(x) = (2x^2-2)\,e^{2x^2-1}$

$$\begin{cases} y = (2x^2-2)\,e^{2x^2-1} \\ y = 4e^{2x^2-1} \end{cases}$$

$$(2x^2-2)\,e^{2x^2-1} = 4e^{2x^3-1}$$

$$\begin{cases} 2x^2-2 = 4 \\ // \end{cases} \quad \begin{cases} x^2-1 = 2 \\ y_{1,2} = 4e^{2\cdot 3-1} \end{cases} \quad \begin{cases} x^2 = 3 \\ y = 4e^5 \end{cases} \quad \begin{cases} x_{1,2} = \pm\sqrt{3} \\ // \end{cases}$$

$A\left(+\sqrt{3};\ 4e^5\right)$

$B\left(-\sqrt{3};\ 4e^5\right)$

$$\begin{cases} y = (2x^2-2)\,e^{2x^2-1} \\ y = 2x^2-2 \end{cases} \quad (2x^2-2)\,e^{2x^2-1} = 2x^2-2 \quad (2x^2-2)\left(e^{2x^2-1}-1\right) = 0$$

$2x^2-2 = 0 \quad x^2 = 1 \quad ; \quad x = \pm 1$

$e^{2x^2-1} - 1 = 0 \quad e^{2x^2-1} = 1 \quad ; \quad 2x^2 - 1 = 0 \to x^2 = \frac{1}{2} \to x_{3,4} = \pm\sqrt{\frac{1}{2}}$

$C(1,0) \quad D(-1,0)$

$x_{3,4} = \pm\frac{\sqrt{2}}{2}$

$E\left(+\frac{\sqrt{2}}{2}, -1\right) \quad F\left(-\frac{\sqrt{2}}{2}, -1\right)$

$y_{1,2} = 2(\pm 1)^2 - 2 = 0$

$y_{3,4} = 2\left(\pm\frac{1}{\sqrt{2}}\right)^2 - 2 = \frac{2}{2} - 2 = -1$

978 Considera la funzione $f(x) = 2\log_2(a + |x-1|) - 4$, con $a \in \mathbb{R}$.

a. Determina per quali valori di a il suo dominio è $\mathbb{R}$.

b. Trova per quale valore di a risulta $f(2) = 0$.

c. Traccia il grafico della funzione ottenuta in corrispondenza del valore di a trovato.

d. Scrivi l'equazione $y = g(x)$ della retta passante per i due punti del grafico di $y = f(x)$ di ascisse 2 e 6.

e. Risolvi graficamente la disequazione $f(x) \geq g(x)$.

[a. $a > 0$; b. $a = 3$; c. $y = \frac{1}{2}x - 1$; d. $x \leq \alpha \vee 2 \leq x \leq 6$, con $\alpha \in (0, 1)$]

- $f(x) = 2\log_2(a + |x-1|) - 4$ $\quad D: \mathbb{R}$

$a + |x-1| > 0$

$a > -|x-1|$ a deve essere maggiore di TUTTO l'insieme che va da $-\infty$ a 0, quindi a deve essere > 0

$\boxed{a > 0}$

- $f(2) = 0$

$0 = 2\log_2(a + |2-1|) - 4$

$0 = 2\log_2(a+1) - 4$

$2\log_2(a+1) = 4$

$\log_2(a+1) = 2$

$2 = a + 1$

$a + 1 = 4$

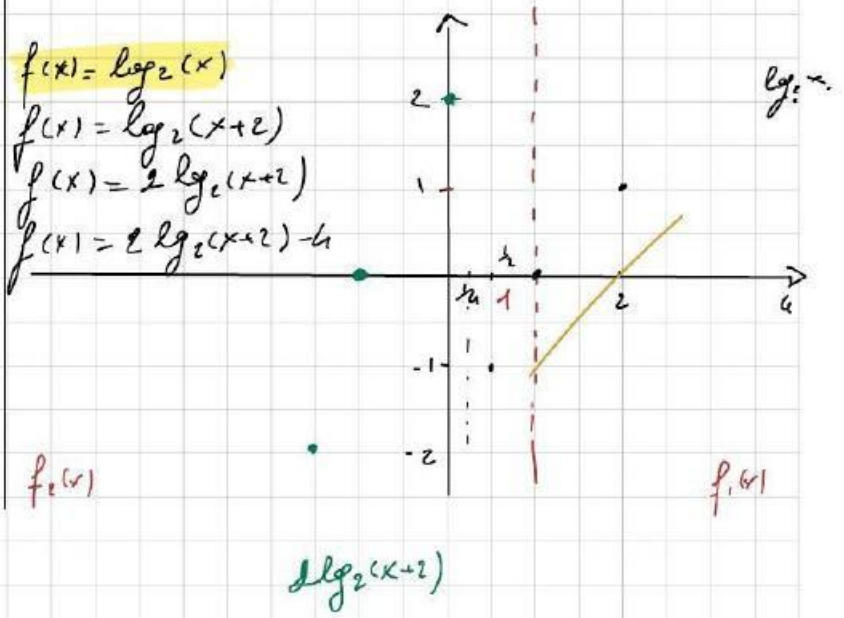
$a = 3$

- $f(x) = 2\log_2(|x-1| + 3) - 4$

$|x-1| = \begin{cases} x-1 & \text{se } x \geq 1 \\ 1-x & \text{se } x < 1 \end{cases}$

$f_1(x) = 2\log_2(x+2) - 4$ se $x \geq 1$

$f_2(x) = 2\log_2(4-x) - 4$ se $x < 1$

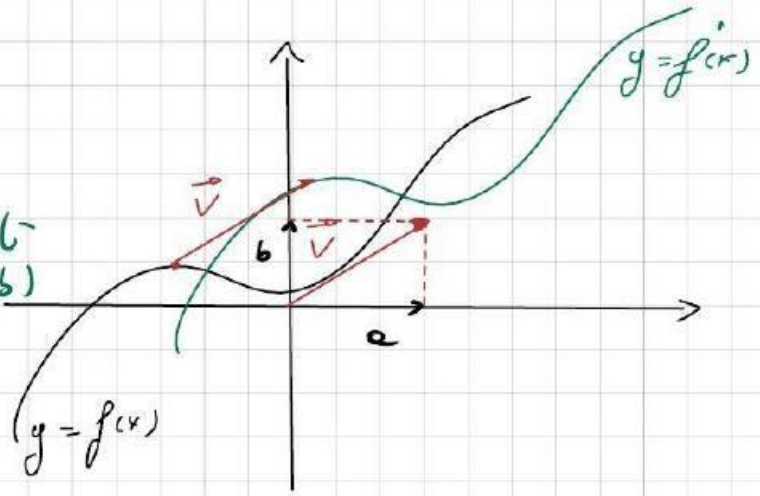


$2\log_2(x+2)$

TRASLAZIONI

$y = f(x)$ $\quad y = f(x-a)+b$

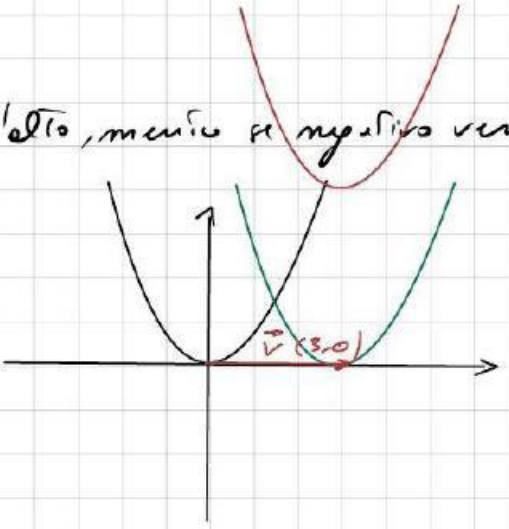
$y - b = f(x-a)$ Traslazione di un vettore $\vec{v}(a,b)$



- Se $a > 0$, la funzione si sposta verso dx, mentre se negativo, verso sx.
- Se $b > 0$, la funzione si sposta verso l'alto, mentre se negativo verso il basso.

Esempi:

$y = (x-3)^2$ Traslazione della funzione $y = x^2$.



$y = (x-3)^2 + 2$

$y = f(x+2) + 3$ (← ↑)

$y = f(x-2) + 3$ (→ ↑)

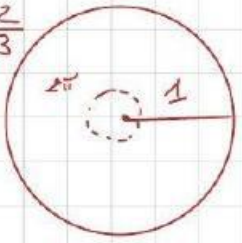
$y = f(x-2) - 3$ (→ ↓)

$y = f(x+2) - 3$ (← ↓)

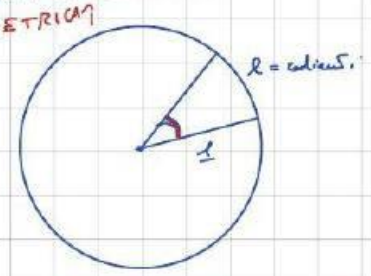
I RADIANTI

I radianti sono un altro modo di rappresentare gli angoli. Al posto di 360° come angolo piatto, si usa la lunghezza della circonferenza di raggio 1.

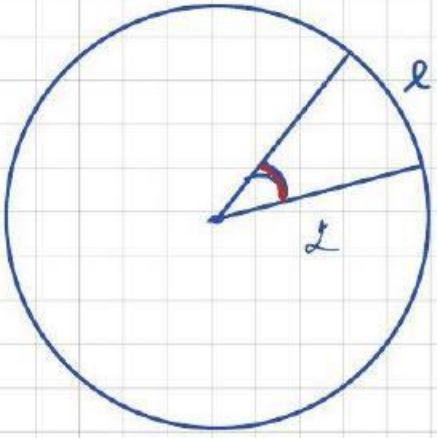
$\alpha = 360° = 2\pi \cdot 1 = 2\pi = \underbrace{6,28}_{(rad)} (rad) \simeq 2 \cdot \frac{22}{7}$



CIRCONFERENZA GONIOMETRICA



l'angolo in radianti è l'arco di circonferenza che delimita l'angolo se il raggio vale 1.



$$\alpha = \frac{l}{r}$$

EQUIVALENZE

- $21,347° = 21°20,82' = 21°20'49''$

grado primo secondo

$0,347 \times 60 = 20,82$

$0,82 \times 60 = 49,2$

- $45°12'2'' = 45° + 12 \cdot \frac{1°}{60} + 2 \cdot \frac{1}{3600}° = 45,20056$

$1' = \frac{1}{60}°$; $1'' = \frac{1}{60}' = \left(\frac{1}{60}\frac{1}{60}\right)° = \frac{1}{3600}°$

CONVERSIONI TRA ° e Rad

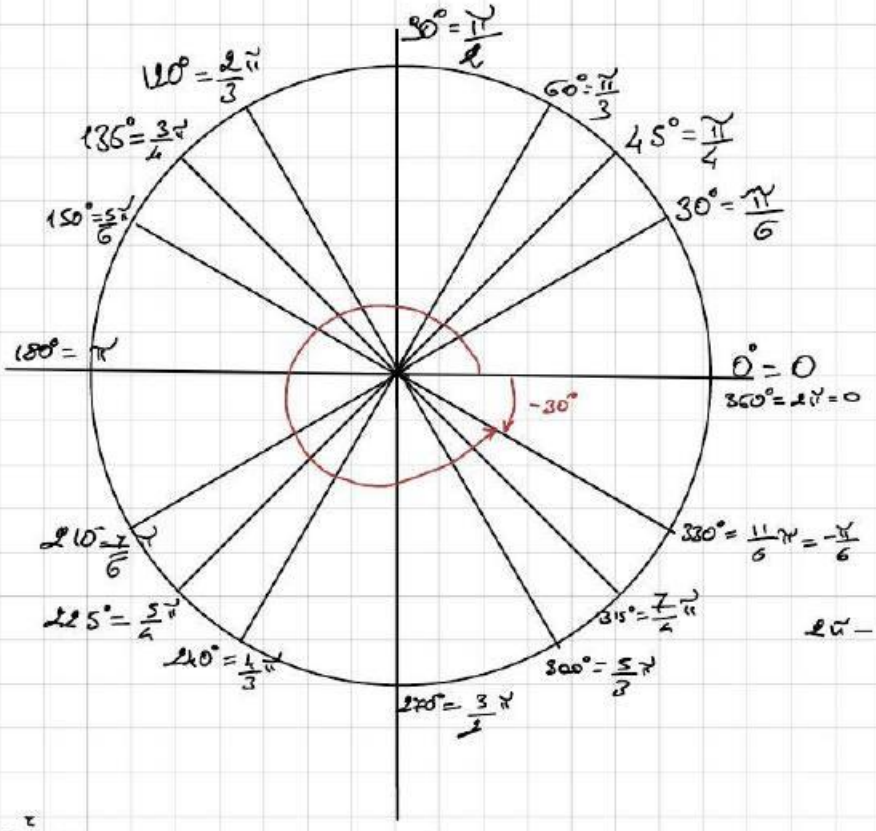
$\alpha° : 180° = \alpha^r : \pi$

$\alpha° = \alpha^r \cdot \frac{180°}{\pi}$

$\alpha^r = \alpha° \cdot \frac{\pi}{180}$

$20° = \frac{\pi}{9}$ rad

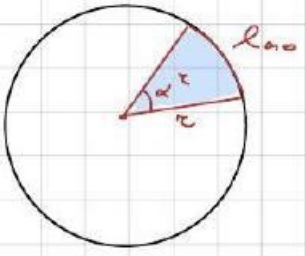
$\alpha^r = 20° \cdot \frac{\pi}{180°} = \frac{\pi}{9}$



$2\pi - \frac{\pi}{6} = \frac{11}{6}\pi$

$l_{arco} = \alpha^r \cdot r$

$A_{settore\ circolare} = \frac{1}{2} r \cdot l = \frac{1}{2} r^2 \alpha^r$



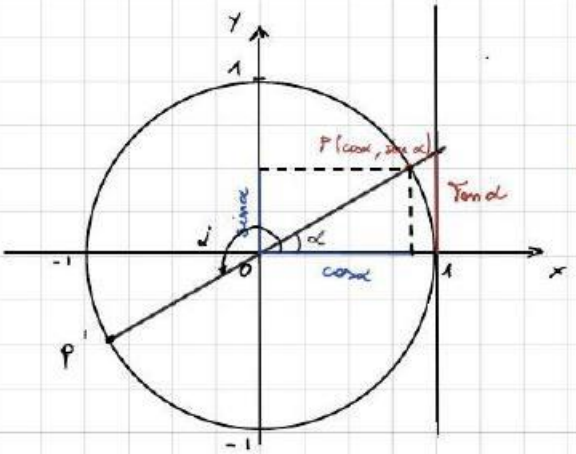
FUNZIONI GONIOMETRICHE

	sen	cos
I quadrante :	+	+
II " :	+	−
III " :	−	−
IV " :	−	+

$\alpha' = 180 + \alpha$

$\alpha' - \alpha = 180$

$\tan\alpha = \tan\alpha'$



$\tan\alpha : 1 = \sin\alpha : \cos\alpha$

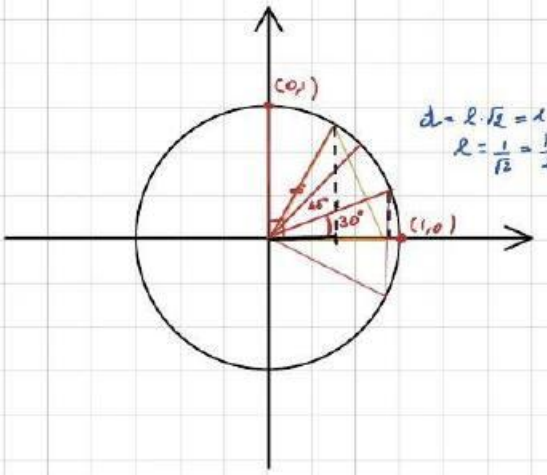
$\tan\alpha = \frac{\sin\alpha}{\cos\alpha}$

Chiamiamo:

- seno di α l'ordinata di P;
- coseno di α l'ascissa di P;
- tangente di α il rapporto tra l'ordinata e l'ascisse di P.

$$\tan\alpha = \frac{\sin\alpha}{\cos\alpha}$$

VALORI NOTEVOLI F. G. DI ANGOLI NOTI



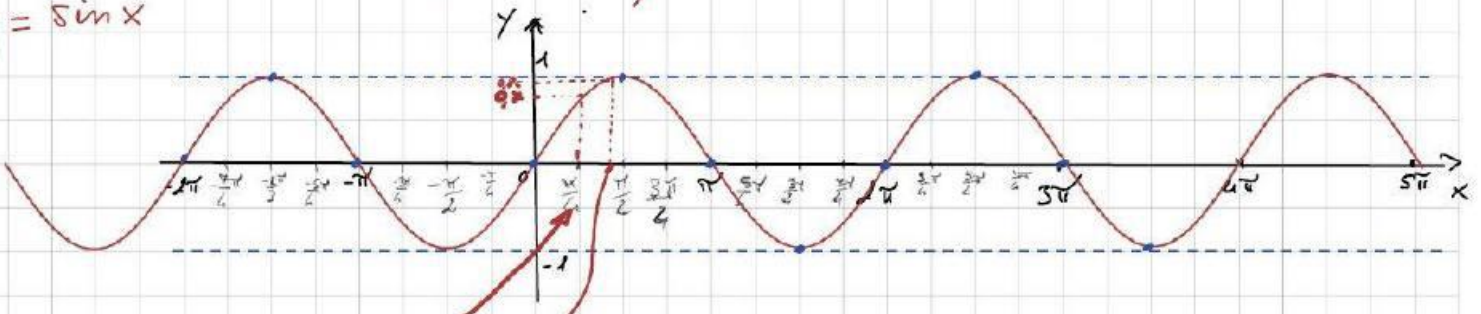
α^c	α°	sin	cos	tan
0	$0°$	0	1	0
$\pi/6$	$30°$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$\pi/4$	$45°$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\pi/3$	$60°$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	$90°$	1	0	$\exists$

Tecnica mnemonica.

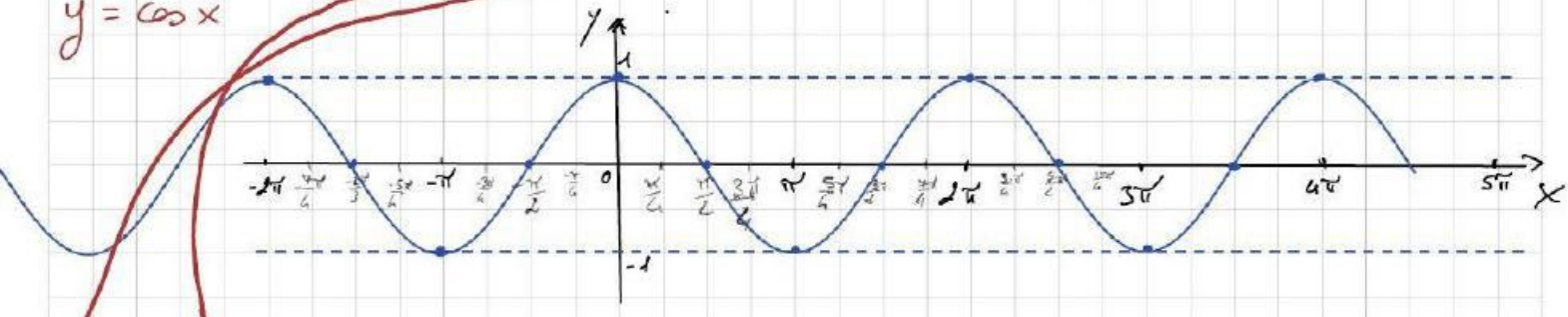
	0	30°	45°	60°	90°
sen	$\frac{\sqrt{0}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{4}}{2}$
cos	$\frac{\sqrt{4}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{0}}{2}$

FUNZIONE SENO, COSENO E TANGENTE

$y = \sin x$



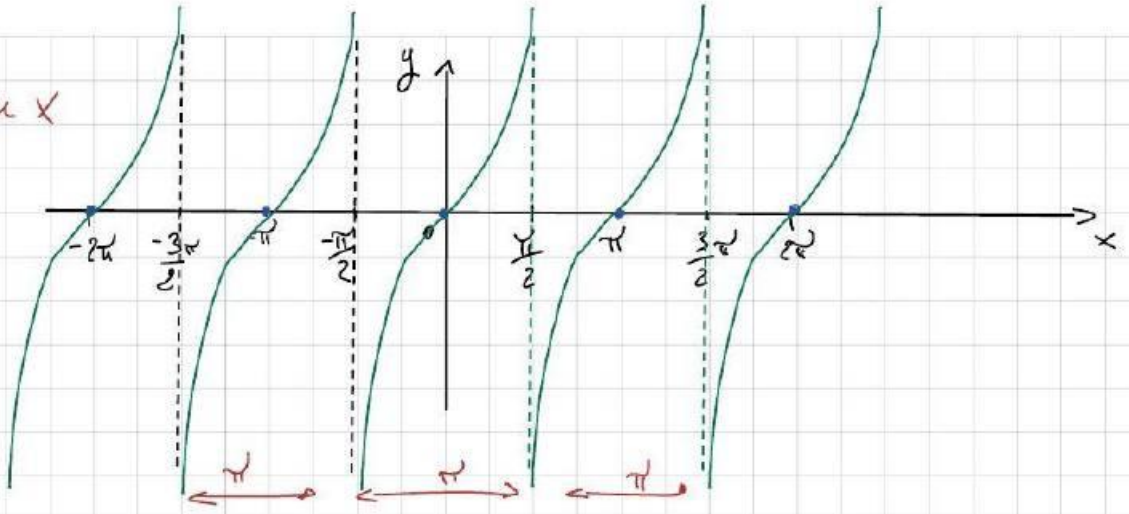
$y = \cos x$



$\sin 45° = \frac{\sqrt{2}}{2} = 0,7071\ldots$

$\sin 60° = \frac{\sqrt{3}}{2} = 0,866$

$y = \tan x$



$$\frac{\log_3(x^2-2x)-1}{\log_{\frac{1}{3}}(x-2)+1} \geq 0$$

C.E.

$$\begin{cases} x^2-2x>0 \\ x-2>0 \end{cases} \quad \begin{cases} x(x-2)>0 \\ x>2 \end{cases}$$

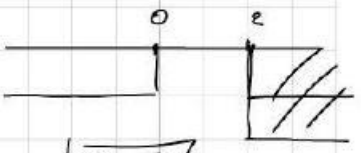
N: $\log_3(x^2-2x)-1 \geq 0$

$\log_3(x^2-2x) \geq \log_3 3$

$x^2-2x \geq 3 \rightarrow x^2-2x-3 \geq 0 \rightarrow x \leq -3 \vee x \geq 1$

$x<0 \vee x>2$

$x>2$

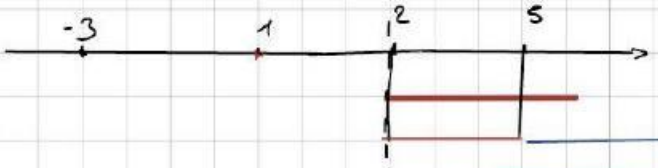


D: $\log_{\frac{1}{3}}(x-2) > -1$

$\log_{\frac{1}{3}}(x-2) < \log_{\frac{1}{3}}\left(\frac{1}{3}\right)^{-1}$

$x-2<3 \rightarrow x<5$

$x>2$

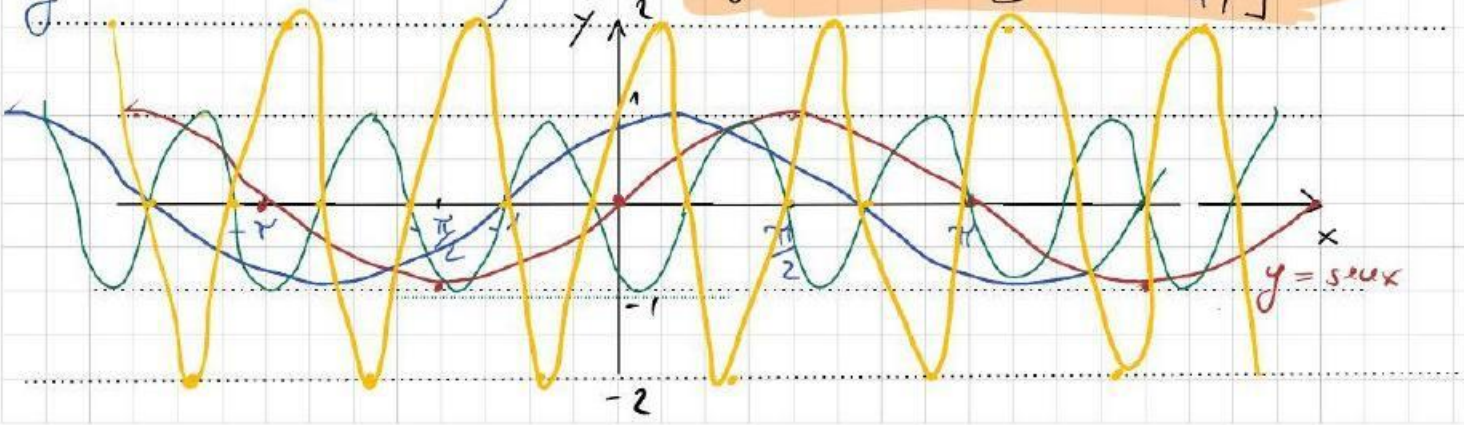


$2 < x < 5$

TRASF. GEOMETRICHE

$y = -2\,\text{sen}(4x+1)$

$y = -2\,\text{sen}\left[4\left(x+\frac{1}{4}\right)\right]$



1) $y = \sin x$

2) $y = \sin\left(x + \frac{1}{a}\right)$

3) $y = \sin\left[a\left(x + \frac{1}{a}\right)\right]$

4) $y = -2 \sin\left[a\left(x + \frac{1}{a}\right)\right]$

funzione elementare

appl. la traslazione $\frac{1}{a}$ ←

Restring. lungo x di un fattore a

Dilatazione e ribalt. lungo l'asse Y

$y = a \cdot \sin[b \cdot (x + c)] + d$

Es: $y = 2 \sin[3(x - 4)] + 5$

Traslazione lungo y
$d > 0$, Tras ↑
$d < 0$, Tras ↓

Traslazione lungo x
$c > 0$, Tras. ←
$c < 0$, Tras →

dilatazione lungo x
$b > 1$ accorciamento →←
$0 < b < 1$ allungamento ← →
$b = +1$ nessuna dilatazione
$b = -1$ "ribalta" l'asse x.

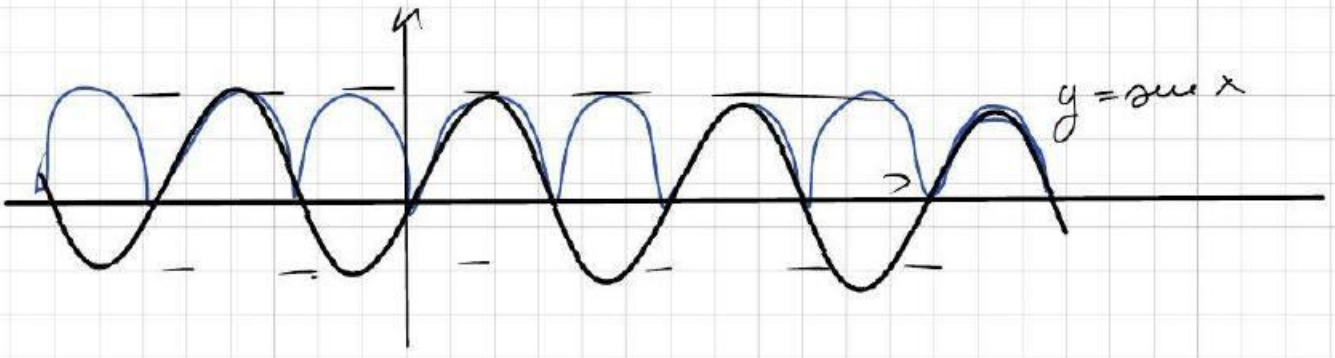
dilatazione lungo y
$a > 1$ allungamento ↑↓
$0 < a < 1$ accorciamento ↓↑
$a = 1$ nessuna dilatazione
$a = -1$ ribalta l'asse y.

$y = \sin x$

$y = \sin |x|$

$y = |\sin x|$

$y = 3 \sin(|x| + 2)$ il valore assoluto si applica alla fun.



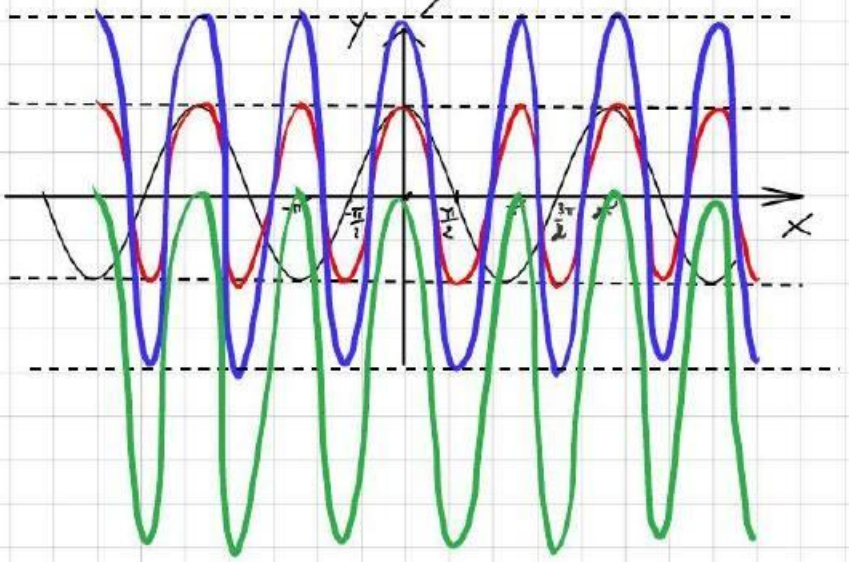
371 $y = 2\cos 2x - 2$

$y = \cos x$

$y = \cos 2x$ dilat. lungo asse x

$y = 2\cos 2x$ " " " y

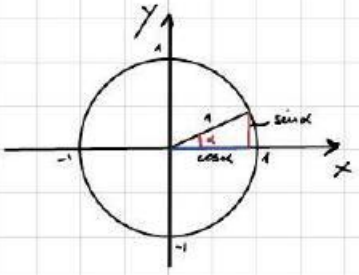
$y = 2\cos 2x - 2$ Traslazione ↓

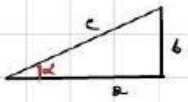


$y = \cos(2x - 4)$

$y = \cos[2(x-2)]$

FUNZIONI GONIOMETRICHE E TRIANGOLI RETTANGOLI





$\cos\alpha : a = 1 : c$

$\sin\alpha : b = 1 : c$

$\cos\alpha = \frac{a}{c}$

$\sin\alpha = \frac{b}{c}$

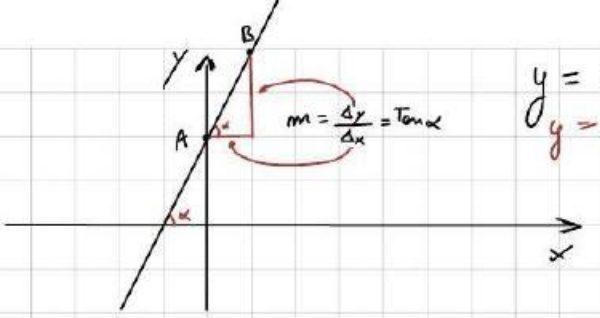
$\sin\alpha = \frac{\text{cateto opposto}}{\text{ipotenusa}} \rightarrow$ cateto opposto = ipotenusa · $\sin\alpha$; $b = c \cdot \sin\alpha$

$\cos\alpha = \frac{\text{cateto adiacente}}{\text{ipotenusa}} \rightarrow$ cateto adiacente = ipotenusa · $\cos\alpha$; $a = c \cdot \cos\alpha$

} I Teorema dei Triangoli rettangoli

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = \frac{\frac{b}{c}}{\frac{a}{c}} = \frac{b}{a} = \frac{\text{cateto opposto}}{\text{cateto adiacente}}$

} II Teorema dei Triangoli rettangoli

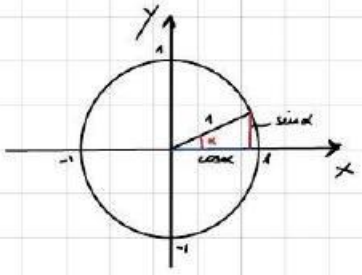


$m = \frac{\Delta y}{\Delta x} = \tan\alpha$

$y = mx + q$ $\quad m = 2$

$y = 2x + 2$ $\quad q = 2$

IDENTITÀ GEOMETRICA FONDAMENTALE



Applicazione del T. di P.

$$\sin^2\alpha + \cos^2\alpha = 1$$

$$\sin^2\alpha = 1 - \cos^2\alpha$$

$$\sin\alpha = \pm\sqrt{1-\cos^2\alpha}$$

$$\cos\alpha = \pm\sqrt{1-\sin^2\alpha}$$

Calcoliamo il sin o cos della Tangente

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha}$

$$\frac{1}{\cos^2\alpha} = \frac{\sin^2\alpha + \cos^2\alpha}{\cos^2\alpha} = \frac{\sin^2\alpha}{\cos^2\alpha} + 1 = \tan^2\alpha + 1$$

$$\frac{1}{\cos^2\alpha} = \tan^2\alpha + 1 \longrightarrow \cos^2\alpha = \frac{1}{1+\tan^2\alpha} \longrightarrow \boxed{\cos\alpha = \pm\frac{1}{\sqrt{1+\tan^2\alpha}}}$$

In maniera analoga:

$$\boxed{\sin\alpha = \pm\frac{|\tan\alpha|}{\sqrt{1+\tan^2\alpha}}}$$

236 $1 - \cos^2\theta = \dfrac{\tan^2\theta}{1+\tan^2\theta}$

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha}$

C.E. $1 + \tan^2\alpha \neq 0$

$$\frac{\tan^2\alpha}{1+\tan^2\alpha} = \frac{\frac{\sin^2\alpha}{\cos^2\alpha}}{1+\frac{\sin^2\alpha}{\cos^2\alpha}} = \frac{\frac{\sin^2\alpha}{\cos^2\alpha}}{\frac{\cos^2\alpha+\sin^2\alpha}{\cos^2\alpha}} = \frac{\frac{\sin^2\alpha}{\cos^2\alpha}}{\frac{1}{\cos^2\alpha}} =$$

$$= \frac{\sin^2\alpha}{\cos^2\alpha}\cdot\cos^2\alpha = \sin^2\alpha$$

$$1 - \cos^2\alpha = \sin^2\alpha$$

$$1 - \cos^2\alpha = 1 - \cos^2\alpha.$$

$$\sin^2\alpha + \cos^2\alpha = 1$$

$$\sin^2\alpha = 1 - \cos^2\alpha$$

237 $\sin^2\alpha \tan^2\alpha = \tan^2\alpha - \sin^2\alpha$

$$\sin^2\alpha \cdot \frac{\sin^2\alpha}{\cos^2\alpha} = \frac{\sin^2\alpha}{\cos^2\alpha} - \sin^2\alpha$$

$$\frac{\sin^2\alpha \cdot \sin^2\alpha}{\cos^2\alpha} = \frac{\sin^2\alpha - \sin^2\alpha\cos^2\alpha}{\cos^2\alpha}$$

$$\frac{\sin^4\alpha}{\cos^2\alpha} = \frac{\sin^2\alpha\left(1-\cos^2\alpha\right)}{\cos^2\alpha}$$

$$\frac{\sin^4\alpha}{\cos^2\alpha} = \frac{\sin^2\alpha \cdot \sin^2\alpha}{\cos^2\alpha}$$

$$\frac{\sin^4\alpha}{\cos^2\alpha} = \frac{\sin^4\alpha}{\cos^2\alpha}$$

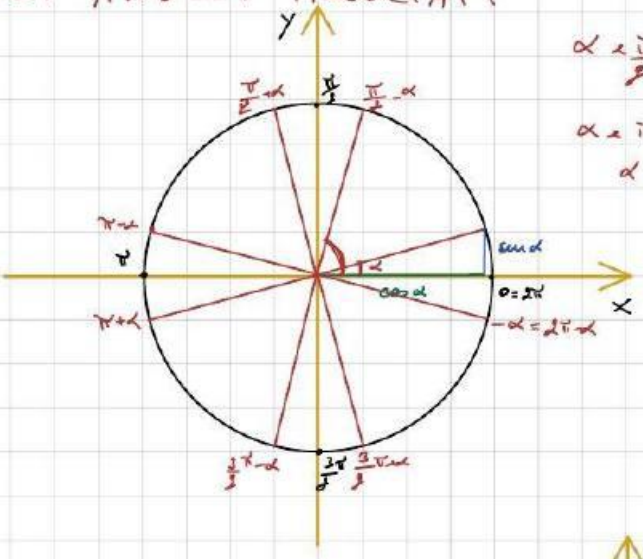
$\sin^2\alpha + \cos^2\alpha = 1$

$1 - \cos^2\alpha = \sin^2\alpha$

$1 - \sin^2\alpha = \cos^2\alpha$

$\frac{\sin^2\alpha}{\sin^2\alpha / \cos^2\alpha} = \sin^2\alpha \cdot \tan^2\alpha$

GLI ANGOLI ASSOCIATI



α e $\frac{\pi}{2}-\alpha$ sono complementari

α e $\pi-\alpha$ sono supplementari

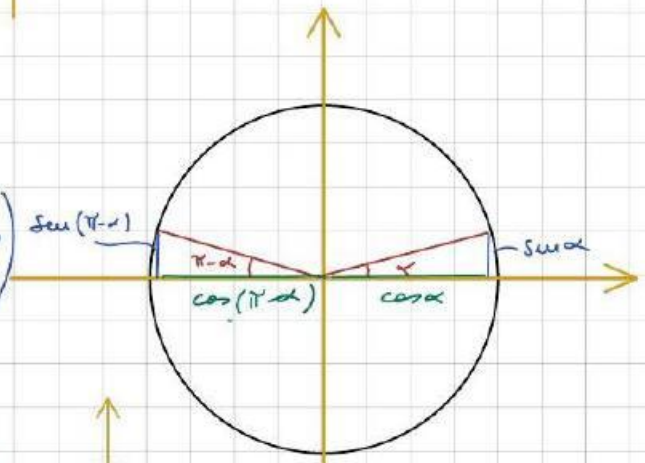
α e $-\alpha$ sono opposti

ANGOLI SUPPLEMENTARI

$\sin(\pi-\alpha) = \sin\alpha$

(Es $\sin 170 = \sin(180-10) = \sin 10$)

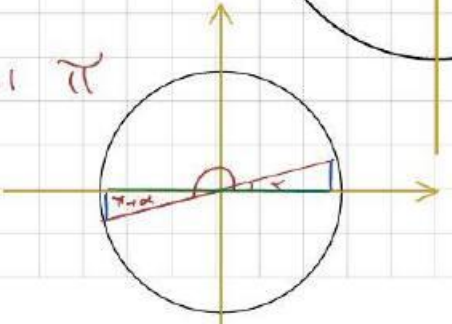
$\cos(\pi-\alpha) = -\cos\alpha$



ANGOLI CHE DIFFERISCONO DI π $(\alpha, \pi+\alpha)$

$\sin(\pi+\alpha) = -\sin\alpha$

$\cos(\pi+\alpha) = -\cos\alpha$



ANGOLI COMPLEMENTARI $\left(\alpha, \frac{\pi}{2} - \alpha\right)$

$\cos\left(\frac{\pi}{2} - \alpha\right) = \sin\alpha$

$\sin\left(\frac{\pi}{2} - \alpha\right) = \cos\alpha$

ANGOLI CHE DIFF. DI $\frac{\pi}{2}$ $\left(\alpha, \frac{\pi}{2} + \alpha\right)$

$\cos\left(\frac{\pi}{2} + \alpha\right) = -\sin\alpha$

$\sin\left(\frac{\pi}{2} + \alpha\right) = \cos\alpha$

